the $n(d)$ different Φs that transform under the gauge group according to the representation d. The only way to construct a function of the Φs that is invariant under the global symmetry group $\prod_d SU(n(d))$ is as a product of Φs, with the $n(d)$ flavor indices contracted for each d with the antisymmetric $SU(n(d))$ tensor $\epsilon_{i_1\ldots i_{n(d)}}$, and with the gauge indices contracted with constant tensors of the gauge group. (For instance, for generalized supersymmetric quantum chromodynamics, v_λ must be a function of the sole invariant

$$D \equiv \mathrm{Det}_{ij} \sum_a Q_{ai}\overline{Q}_{aj} \,,$$

which is non-zero only for $N_c \geq N_f$.)

In addition to the anomaly-free $SU(n(d))$ flavor symmetries, there is also a $U_d(1)$ symmetry for each of the irreducible representations d furnished by the Φs, with all $\Phi^{(d)}_{ai}$ for a given d undergoing the transformation

$$\Phi^{(d)}_{ai} \to e^{i\varphi_d}\Phi^{(d)}_{ai} \,. \tag{29.3.17}$$

This symmetry is anomalous, with the effects of the anomaly the same as if the Lagrangian underwent the change

$$\mathscr{L} \to \mathscr{L} - \sum_d \frac{n(d)C_{2d}}{32\pi^2} \sum_A \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma} \varphi_d \,, \tag{29.3.18}$$

where C_{2d} is the contribution to C_2 of any one left-chiral scalar superfield belonging to the irreducible representation d of the gauge group. The symmetry is restored if we give T the transformation property

$$T \to T + n(d)C_{2d}\varphi_d/\pi \,. \tag{29.3.19}$$

Since $v_\lambda(\Phi)$ is accompanied in Eq. (29.3.9) by a factor $\exp(2i\pi T/(C_1 - C_2))$, which undergoes the transformation

$$\exp\left(\frac{2i\pi T}{C_1 - C_2}\right) \to \prod_d \exp\left(\frac{+2in(d)C_{2d}\varphi_d}{C_1 - C_2}\right) \exp\left(\frac{2i\pi T}{C_1 - C_2}\right) \,, \tag{29.3.20}$$

we conclude that *for each representation d of the gauge group furnished by the left-chiral scalars, $v_\lambda(\Phi)$ must be a homogeneous function of the $\Phi^{(d)}_{ai}$ of negative order $-2n(d)C_{2d}/(C_1 - C_2)$*. (For instance, in generalized supersymmetric quantum chromodynamics we have two irreducible representations of $SU(N_c)$, the defining and antidefining representations, each with $n(d) = N_f$ and $C_{2d} = 1/2$, so v_λ is a homogeneous function of order $-N_f/(N_c - N_f)$ in the Q belonging to the defining representation and of the same order in the $\bar{Q}$ belonging to the antidefining representation. Thus it must be proportional to $D^{-1/(N_c - N_f)}$, where D is the determinant introduced earlier.) In general $C_2 = \sum_d n(d)C_{2d}$, so $v_\lambda(\Phi)$ is a homogeneous function of all the Φs, of order $-2C_2/(C_1 - C_2)$.